INTEGRATED FINAL PROJECT GUIDELINES

Course 1	Course 2
Software Engineering I (CS 312)	**Intelligent Systems (CSE 103)**
Provides the process, documentation, and quality framework.	Provides the AI technique and the working chatbot to be engineered.

1. Rationale

This guideline merges the two final requirements into a single project: students will design and build one AI chatbot, using a formal Software Engineering process to do it.

Software Engineering I emphasizes the discipline of building software correctly (Process models, requirements, design, testing, quality, and project management)

Intelligent Systems emphasizes AI techniques (search, knowledge representation, neural networks, fuzzy logic, genetic algorithms, expert systems)

Software Engineering 1 Satisfied	Intelligent Systems Satisfied
• Explain and apply SDLC phases • Analyze/critique SE issues: quality, project management, ethics • Write and present an analysis of process models, quality, project management, and design improvements • Work effectively in a group on planning, testing, and final presentation	• Apply ANN, Fuzzy Logic, GA, or Expert System techniques to a practical scenario • Develop a working AI system/chatbot integrating course concepts • Communicate technical ideas through documentation and presentation • Demonstrate awareness of ethical/societal implications of AI

2. Group Formation

- 6–7 members per group.
- Each group develops one AI chatbot for a chosen domain or audience (ex. school services/FAQ, health advice, tourism, customer support, etc.) and must incorporate at least one of: Artificial Neural Network, Fuzzy Logic, Genetic Algorithm, or Expert System rules.
- Each group selects one SDLC/process model; Waterfall Model, Spiral Model, V-Model, RAD Model or Agile Model.
- Groups must assign clear roles (Project Manager, Lead Developer, AI/ML Developer, QA/Tester, Documentation Lead, etc.), to be stated in the Project Management Plan.

2.1 Build-From-Scratch Rule (No External AI APIs)

Build-From-Scratch Rule (No External AI APIs)
The chatbot's intelligence must be designed, coded, and trained by the group itself. Students may not outsource the AI component to an online service.

- NOT ALLOWED: calling online AI/chatbot APIs or services (e.g., OpenAI/ChatGPT, Google Gemini, Claude, Dialogflow, Watson, Microsoft Bot Framework/LUIS, Hugging Face hosted inference, or any similar pre-trained cloud model or API).
- NOT ALLOWED: downloading and simply plugging in a ready-made pre-trained chatbot or language model as the chatbot's "brain."
- REQUIRED: the group creates its own dataset or knowledge base (intents, patterns, responses, or rules) specific to their chosen domain.
- REQUIRED: the group trains or builds the AI component itself — for example, a self-trained neural network for intent classification, a hand-written fuzzy inference system, a genetic algorithm they implemented, or a rule-based expert system with their own rule base.
- ALLOWED: general programming languages and standard libraries for building and training their own model (e.g., Python, NumPy, TensorFlow/Keras, PyTorch, scikit-learn, NLTK), and a plain interface (web page, desktop app, or console). These are tools for building, not ready-made AI services.
- The chatbot must run without depending on a paid or online AI service to produce its answers.

2.2 Evidence of Original Work

To show the AI component is the group's own work, each group must submit and be ready to explain:

- The dataset/knowledge base they built (with the source/documentation method of collection).
- The training code or rule base, with a version-control history (e.g., GitHub commits) showing contributions from members.
- Training results: parameters used, training/testing split, and accuracy or error results.
- A code walkthrough during the defense. Any member may be asked to explain how the AI component works.

3. Timeline

Mapped directly onto the existing weekly schedule - no adjustment

Week	Timeline	Deliverable
13	Groups formed	Group Members + preferred chatbot domain (informal)
14	Proposal drafting	Draft SRS (functional/non-functional requirements)
15	**OFFICIAL LAUNCH (both subjects)**	**Approved Project Proposal: chatbot domain, chosen AI technique, chosen SDLC model, justification**
16	Design & Planning	System Design doc (architecture, diagrams) + Project Management Plan (Gantt, roles, risk register)
17	Build sprint 1 and 2 + Testing	Working core chatbot features + progress log Test Plan & Test Cases executed; Quality metrics evaluation; Ethics reflection
18	**FINAL DEFENSE (both subjects)**	**Live chatbot demo + full IEEE paper (Overleaf) + oral defense**

4. Weekly Checklist

Phase	SE1 Requirement	IS Requirement
Proposal	Chosen SDLC model + rationale	Chatbot domain + AI technique (ANN/Fuzzy/GA/Expert System)
Requirements	SRS: functional & non-functional requirements	Knowledge base / training data and conversation requirements
Design	Architecture diagram, design pattern(s) used, quality attributes	Chatbot conversation flow + AI design (network structure, fuzzy rules, GA parameters, or rule base)
Implementation	Coding standards, version control log	Working chatbot with the AI component integrated
Testing	Test plan, test cases, unit/integration/system/acceptance results	Response accuracy/performance evaluation of the chatbot
Project Mgmt	Gantt chart, roles, risk register	— (shared artifact)
Quality & Ethics	Quality metrics assessment	Ethical/societal impact reflection
Final Paper	Process model case study (replaces external research paper analysis)	Technical documentation of the chatbot's AI approach

Phase	SE1 Requirement	IS Requirement
Defense	Defend process, design, and quality decisions	Live chatbot demo + defend AI technique and results
Phase	SE1 Requirement	IS Requirement

Checklist

Proposal		
Requirements		
Design		
Implementation		
Testing		
Project Mgmt		
Quality & Ethics		
Final Paper		
Defense		

5. Final Paper Format — IEEE via Overleaf

All groups will write their final documentation as a single IEEE-format conference paper, prepared in Overleaf using the standard IEEEtran two-column template.

Overleaf setup

- Create a new project in Overleaf → search template “IEEE Conference Template
- One Overleaf project per group; add all members as collaborators/editors for shared editing.
- Submit the Overleaf share link (read/edit) and a exported PDF on or before the Final Presentation

Required sections

Section	Content Focus
Abstract & Keywords	Summary of the chatbot, AI technique used, and process model followed (150–250 words)
I. Introduction	Problem/context, objectives, scope of the chatbot
II. Related Work	Brief review of similar chatbots or AI techniques (with IEEE-style citations)
III. Methodology	Chosen SDLC model, justification, and how each phase was carried out
IV. System Design	Architecture diagrams, design patterns, AI model/algorithm design
V. Implementation	Tools, technologies, and how the AI component was built
VI. Testing & Results	Test cases, results, chatbot response accuracy metrics
VII. Discussion	Software quality assessment, risks encountered, ethical considerations
VIII. Conclusion	Summary, limitations, recommendations for future work
References	IEEE citation style, numbered in order of appearance

Target length: 6–10 pages in the two-column IEEE format, excluding references.

6. Final Defense Format

- **TARGET DATE: NOVEMBER 16, 2026 (TENTATIVE)**
- One unified defense session per group (no separate SE1 and IS defenses).
- Suggested duration: 20 minutes presentation + 10 minutes open panel Q&A.
- Live demo of the working chatbot is required, running on the group’s own trained model (no external AI API).

- Panel may ask any member to walk through the dataset, training code, or rule base.
- A single panel evaluates using two rubric sheets (SE1 rubric and IS rubric, Section 8) — producing two separate grades from one presentation.

7. Final Defense Rubric

7.1 Software Engineering 1 Rubric (Final Grade component)

Criteria	Weight	Description
Process Model Application	25%	Appropriateness and correctness of the chosen SDLC model and how faithfully it was followed
Requirements & Design	20%	Clarity of SRS, quality of architecture/design diagrams
Testing & Quality	20%	Completeness of test plan/cases, quality metrics used
Project Management	15%	Gantt chart, risk register, evidence of teamwork/role delegation
Paper & Ethics Reflection	10%	IEEE paper quality, depth of process/ethics reflection
Oral Defense	10%	Clarity and command of SE concepts during Q&A

7.2 Intelligent Systems Rubric (Final Grade component)

Criteria	Weight	Description
AI Technique Correctness	30%	Correct use of a self-built and self-trained ANN, Fuzzy Logic, GA, or Expert System (no external AI API)
Functionality / Demo	25%	Chatbot works as intended without online AI services; smooth live demonstration
Performance Evaluation	15%	Response accuracy, efficiency, or quality metrics of the chatbot
Technical Documentation	15%	Clarity of how the AI approach is explained in the paper
Ethics & Societal Impact	10%	Depth of reflection on responsible AI use
Oral Defense	5%	Clarity and command of AI concepts during Q&A